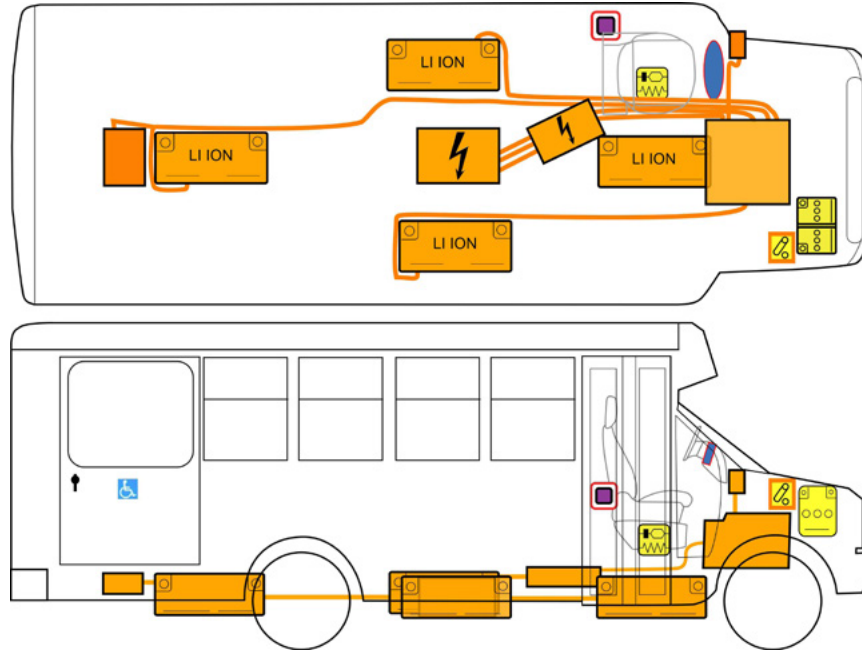




Lightning eMotors Type A School Bus, E-450 Platform 2015-20XX



	Airbag		Battery, low voltage		Seat belt pretensioner		SRS control unit
	Battery, high voltage		High-voltage power cable		Low voltage disconnect to high voltage		Dangerous high voltage

1. Identification / recognition



The electric motor is silent. Vehicle movement exists until vehicle is fully shut down.



Additional Information

Major electrical design is consistent across body variations.

Document Number

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Version

R0

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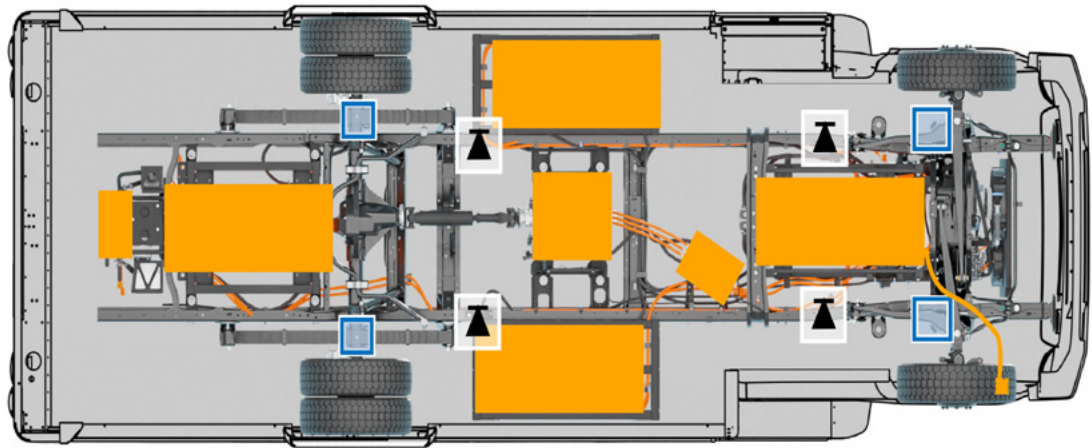
2. Immobilization / stabilization / lifting

Immobilize the vehicle

Apply parking brake
Put vehicle in Park



Lifting points
(on frame only)



Jacking points Lifting points HV components

3. Disable direct hazards / safety regulations



Do not touch, cut, or open high-voltage components and batteries.

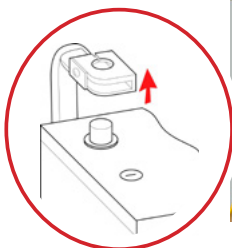
Method 1

Step 1: Turn off ignition.

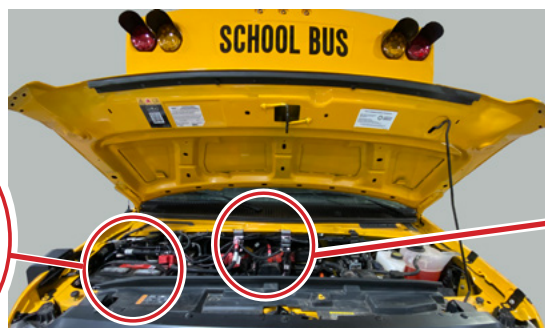


OFF ACCESSORIES RUN

Step 2: Disconnect negative terminal from 12V battery.



Negative terminal



Cut wire location



Lightning eMotors Type A School Bus, E-450 Platform 2015-20XX

Method 3

Step 1: Turn off ignition.

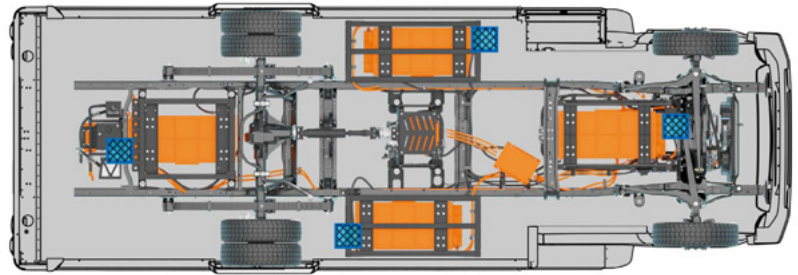
Step 2: Disconnect negative terminal from 12V battery.

Step 3: Remove Manual Service Disconnects (MSD) from all four batteries.

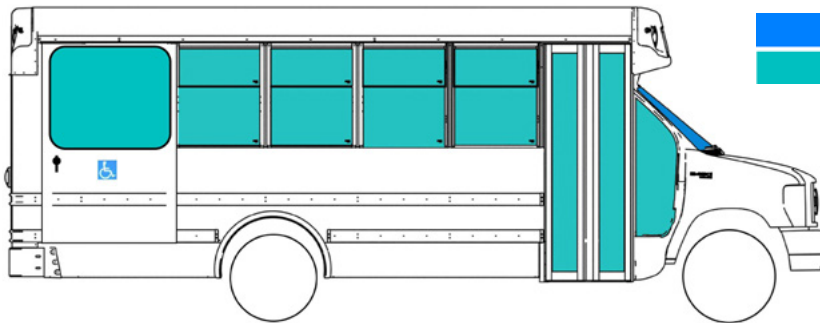


MSD

MSD location



4. Access to occupants



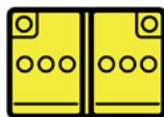
Blue square icon: Laminated safety glass
Cyan square icon: Tempered glass

5. Stored energy / liquids / gasses / solids

350V



12V



If coolant leaks from the battery, there is a risk of a thermal reaction in the high-voltage battery.
Monitor temperature of the battery using an infrared camera.



6. In case of fire

Use copious amounts of water directly on battery to extinguish fire.



Damaged lithium-ion batteries may re-ignite after the fire is extinguished.
Monitor temperature of the battery using an infrared camera.





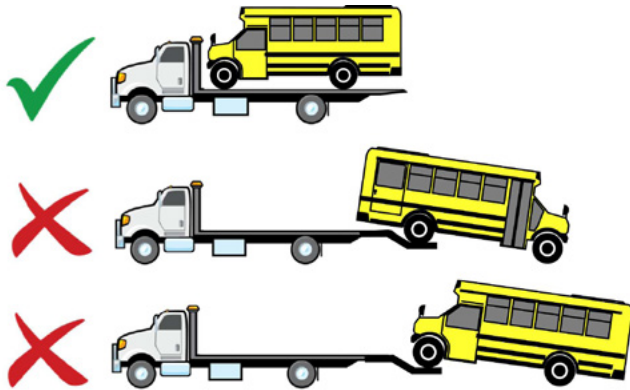
7. In case of submersion

If possible, turn the ignition off in a submerged vehicle to disable the high-voltage system.

No other disabling activities are to be attempted.

Immobilize the vehicle, allow it to dry, and then perform disabling steps in Section 3.

8. Towing / transportation / storage



1. Deactivate high-voltage system (see Section 3).
2. Use infrared camera to confirm high-voltage battery temperature is not above ambient temperature.
3. Park vehicle outdoors at a safe distance from building or other vehicles.



Damaged lithium-ion batteries may re-ignite after the fire is extinguished.
Use infrared camera to check temperature is not above ambient before towing.



9. Important additional information

10. Explanation of pictograms used



Flammable



Explosive



Hazardous to
human health



Environmental
hazard



Acute toxicity



General
warning



Use lots of
water to
extinguish fire



Lithium-ion
battery



Corrosive



High-voltage
warning



Use infrared
camera



Lifting
points